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METHODS AND SYSTEMS FOR REINFORCING ADDITIVELY MANUFACTURED STRUCTURES

Abstract

A method for reinforcing an additively manufactured structure includes draping a textile substrate including one or more conductive fibers over at least a portion of a surface of the additively manufactured structure. The additively manufactured structure includes a thermoplastic matrix material. An electric current is applied to a conductive fiber to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the textile substrate with the thermoplastic matrix material to form a thermoplastic composite on the surface. A system for reinforcing an additively manufactured structure includes a fabrication tool set, an electrical power source and a control device. The control device and the electrical power source apply the electric current to the conductive fiber to heat the thermoplastic matrix material for consolidation of the textile substrate with the thermoplastic matrix material to form a thermoplastic composite.

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Background/Summary

PRIORITY [0001] This application claims priority from U.S. Ser. No. 63/561,374 filed on Mar. 5, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates generally to fiber reinforcement of additively manufactured structures and, particularly, to adding a thermoplastic composite on a surface of additively manufactured structures. Various arrangements of one or more thermoplastic composites on the surface of additively manufactured structures. For example, pieces of textile substrates can be arranged on the surface based on shapes and contours of the additively manufactured structure. The textile substrates are consolidated with a thermoplastic matrix material of the additively manufactured structure to form the thermoplastic composites. The resulting fiber-reinforced additively manufactured structures have numerous applications in aircraft interiors. Additionally, the fiber-reinforced additively manufactured structures have applications as construction materials and components of system equipment and assemblies.

BACKGROUND

[0003] The structural integrity of complex additively manufactured structures, including three-dimensional printed parts, exhibit inherent weakness related to poor layer-to-layer adhesion. Accordingly, those skilled in the art continue with research and development efforts to provide techniques for strengthening and reinforcing additively manufactured structures.

SUMMARY

[0004] Disclosed are examples of methods and systems for reinforcing an additively manufactured structure. The following is a non-exhaustive list of examples, which may or may not be claimed, of the subject matter according to the present disclosure.

[0005] In an example, the disclosed method for reinforcing an additively manufactured structure includes: (1) draping a textile substrate that includes one or more conductive fibers over at least a portion of a surface of the additively manufactured structure, the additively manufactured structure including a thermoplastic matrix material; and (2) applying an electric current to at least one conductive fiber of the one or more conductive fibers to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the textile substrate with the thermoplastic matrix material to form a thermoplastic composite on the surface.

[0006] In another example, the disclosed method for reinforcing an additively manufactured structure includes: (1) adding a first conductive fiber to a fiber reinforcement fabric substrate to form a textile substrate, the first conductive fiber forming a first path distributed about at least a first area of the textile substrate; (2) cutting the textile substrate to form a tailored fabric substrate that fits a surface of the additively manufactured structure, the additively manufactured structure including a thermoplastic matrix material; (3) wrapping the tailored fabric substrate around the surface; and (4) applying an electric current to the first conductive fiber to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the tailored fabric substrate with the thermoplastic matrix material to form a first thermoplastic composite on the

surface and thereby form a fiber-reinforced additively manufactured structure.

[0007] In an example, the disclosed system for reinforcing an additively manufactured structure includes a fabrication tool set, an electrical power source and a control device. The control device is in operative communication with the electrical power source to selectively apply and remove electric current from one or more conductive fibers of a textile substrate. The textile substrate is configured for draping over at least a portion of a surface of the additively manufactured structure. The additively manufactured structure includes a thermoplastic matrix material. The control device and the electrical power source are configured to apply the electric current to at least one conductive fiber of the one or more conductive fibers to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the textile substrate with the thermoplastic matrix material to form a thermoplastic composite on the surface.

[0008] Other examples of the disclosed methods and systems for reinforcing an additively manufactured structure will become apparent from the following detailed description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a flow diagram of an example of a method for reinforcing an additively manufactured structure;

[0010] FIG. 2A is a perspective view of an example of an additively manufactured structure;

[0011] FIG. 2B is a cross-sectional view of the example of the additively manufactured structure shown in FIG. 2A;

[0012] FIG. 3 is a top view of an example of a textile substrate;

[0013] FIG. 4 is a functional block diagram showing an example of the application of electric current to conductive fibers of the textile substrate of FIG. 3 to form a thermoplastic composite on a portion of the surface of the additively manufactured structure of FIG. 2;

[0014] FIG. 5, in combination with FIG. 1, is a flow diagram of another example of a method for reinforcing an additively manufactured structure;

[0015] FIG. 6, in combination with FIG. 1, is a flow diagram of yet another example of a method for reinforcing an additively manufactured structure;

[0016] FIG. 7 is a top view of an example of a textile fabric sheet marked for cutting one or more textile substrates;

[0017] FIG. 8 is a flow diagram of an example of the draping of a textile substrate in the method for reinforcing an additively manufactured structure of FIG. 1;

[0018] FIG. 9 is a perspective view of an example of a tailored substrate cut to fit a portion of the surface of the additively manufactured structure after tacking the textile substrate to the surface;

[0019] FIG. 10, in combination with FIG. 1, is a flow diagram of still another example of a method for reinforcing an additively manufactured structure;

[0020] FIG. 11, in combination with FIG. 1, is a flow diagram of still yet another example of a method for reinforcing an additively manufactured structure;

[0021] FIG. 12, in combination with FIG. 1, is a flow diagram of another example of a method for reinforcing an additively manufactured structure;

[0022] FIG. 13, in combination with FIG. 1, is a flow diagram of yet another example of a method for reinforcing an additively manufactured structure;

[0023] FIG. 14, in combination with FIG. 1, is a flow diagram of still another example of a method for reinforcing an additively manufactured structure;

[0024] FIG. 15, in combination with FIG. 1, is a flow diagram of still yet another example of a method for reinforcing an additively manufactured structure;

[0025] FIG. **16**, in combination with FIG. **1**, is a flow diagram of another example of a method for reinforcing an additively manufactured structure;

[0026] FIGS. **17A-C** are top views of examples of first, second and third textile substrates;

[0027] FIG. **18** is a functional block diagram showing an example of the application of electric current to conductive fibers of first, second and third textile substrates to form first, second and third thermoplastic composites on portions of the surface of the additively manufactured structure of FIG. **2**;

[0028] FIG. **19**, in combination with FIGS. **1** and **16**, is a flow diagram of yet another example of a method for reinforcing an additively manufactured structure;

[0029] FIG. **20**, in combination with FIG. **1**, is a flow diagram of still another example of a method for reinforcing an additively manufactured structure;

[0030] FIG. **21** is a top view of another example of a textile substrate;

[0031] FIG. **22**, in combination with FIGS. **1** and **20**, is a flow diagram of still yet another example of a method for reinforcing an additively manufactured structure;

[0032] FIG. **23** is a flow diagram showing examples of the adding of the one or more conductive fibers in the method for reinforcing an additively manufactured structure of FIG. **20**;

[0033] FIG. **24** is a top view of an example of a carrier sheet marked for cutting one or more carrier substrates;

[0034] FIG. **25**, in combination with FIGS. **1** and **20**, is a flow diagram of another example of a method for reinforcing an additively manufactured structure;

[0035] FIGS. **26A-C** are top views of additional examples of first, second and third textile substrates;

[0036] FIG. **27** is a functional block diagram showing another example of application of electric current to conductive fibers of first, second and third textile substrates to form first, second and third thermoplastic composites on portions of the surface of the additively manufactured structure of FIG. **2**;

[0037] FIG. **28**, in combination with FIGS. **1**, **20** and **25**, is a flow diagram of yet another example of a method for reinforcing an additively manufactured structure;

[0038] FIG. **29** is a flow diagram of still another example of a method for reinforcing an additively manufactured structure;

[0039] FIG. **30** is a top view of an example of a textile substrate marked for cutting a tailored fabric substrate;

[0040] FIG. **31** is a top view of an example of a tailored fabric substrate of FIG. **30** after being cut from the textile substrate;

[0041] FIG. **32** is a functional block diagram showing an example of application of electric current to first, second and third conductive fibers of the tailored fabric substrate of FIG. **31** to form first, second and third thermoplastic composites on portions of the surface of the additively manufactured structure of FIG. **2**;

[0042] FIG. **33** is a flow diagram showing examples of the adding of the first conductive fiber in the method for reinforcing an additively manufactured structure of FIG. **29**;

[0043] FIG. **34** is a flow diagram of an example of the wrapping of the tailored fabric substrate in the method for reinforcing an additively manufactured structure of FIG. **29**;

[0044] FIG. **35** is a perspective view of an example of a tailored substrate cut to fit a portion of the surface of the additively manufactured structure after tacking the tailored fabric substrate to the surface;

[0045] FIG. **36**, in combination with FIG. **29**, is a flow diagram of still yet another example of a method for reinforcing an additively manufactured structure;

[0046] FIG. **37**, in combination with FIG. **29**, is a flow diagram of another example of a method for reinforcing an additively manufactured structure;

[0047] FIG. **38**, in combination with FIGS. **29** and **37**, is a flow diagram of yet another example of

a method for reinforcing an additively manufactured structure;

[0048] FIG. **39** is a functional block diagram of an example of a system for reinforcing an additively manufactured structure;

[0049] FIG. **40** is a functional block diagram of another example of a system for reinforcing an additively manufactured structure;

[0050] FIG. **41** is a block diagram of aircraft production and service methodology that implements one or more of the examples of methods for reinforcing an additively manufactured structure disclosed herein to produce components of the aircraft; and

[0051] FIG. **42** is a schematic illustration of an aircraft that incorporates one or more examples of components produced using one or more of the examples of methods for reinforcing an additively manufactured structure disclosed herein.

DETAILED DESCRIPTION

[0052] The various examples of methods **100, 500, 600, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1900, 2000, 2200, 2500, 2800, 2900, 3600, 3700, 3800** and systems **3900, 4000** for reinforcing additively manufactured structures **200** disclosed herein identify techniques for creating a thermoplastic composite **404** on a surface **210** of the additively manufactured structure **200**. For example, the techniques establish processes for fabrication of fiber-reinforced three-dimensional printed components for aircraft interiors or building structures. The parts are fabricated through the lamination of tailored fiber placement pre-forms on three-dimensional printed parts. The fiber pre-forms enable structural and artistic patterns to be embedded within plastic surfaces. Lamination can be achieved through passing electric current through the fiber reinforcement directly to generate heat to enable self-lamination into the base thermoplastic materials of the three-dimensional printed component.

[0053] The methods **100, 500, 600, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1900, 2000, 2200, 2500, 2800, 2900, 3600, 3700, 3800** and systems **3900, 4000** disclosed herein increase the structural integrity of complex three-dimensional printed parts, particularly in view of the inherent weakness related to layer adhesion by incorporating carbon fiber. The various techniques remove the need for molds when fabricating fiber composites. Additionally, tailored fiber placement techniques overcome the need for a robotic end effector that limits the geometric size of the end component/structure. The disclosed methods and systems enable highly complex designs and mass customization. The methods and systems pass electrical current through fiber reinforcement (e.g., carbon) directly. The fiber is then directly heated to the temperature required to fuse with the 3D printed material, enabling labor-free joining with structural integrity.

[0054] The various methods and systems disclosed herein involve the combination of fused deposition modeling using three-dimensional printing, tailored fiber placement and electrically heating the tailored fiber placement. The techniques laminate the tailored fiber placement pre-form on three dimensional printed parts via electrical heating to melt the tailored fiber placement into the three-dimensional print. For example, the methods and systems involve laminating the carbon fiber through the direct application of electrical current to heat and melt the fiber into the three-dimensional printed material, rather than using a secondary consolidation process to join the carbon fiber on to the printed part. This includes direct application of current through the tailored fiber placement preforms to join to the three-dimensional printed components.

[0055] Referring generally to FIGS. **1, 2A-B, 3-16, 17A-C, 18-25, 26A-C, 27, 28** and **39**, by way of examples, the present disclosure is directed to a method **100, 500, 600, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1900** for reinforcing an additively manufactured structure **200**. FIG. **1** provides an example of the method **100** for reinforcing an additively manufactured structure **200**. FIG. **2A** provides an example of an additively manufactured structure **200**. FIG. **2B** provides a cross-sectional view of the additively manufactured structure **200**. FIG. **3** provides an example of a textile substrate **300**. FIG. **4** shows an example of the application of electric current **402** to conductive fibers **302** of the textile substrate **300** to form a thermoplastic composite **404** on a

portion **212** of the surface **210** of the additively manufactured structure **200**. FIG. 5, in combination with FIG. 1, provides an example of the method **500** for reinforcing an additively manufactured structure **200**. FIG. 6, in combination with FIG. 1, provides an example of the method **600** for reinforcing an additively manufactured structure **200**.

[0056] FIG. 7 provides an example of a textile fabric sheet **700** marked for cutting one or more textile substrates **300**, **702**, **704**. FIG. 8 provides an example of the draping **102** of a textile substrate in the method **100** for reinforcing an additively manufactured structure **200** of FIG. 1. FIG. 9 provides an example of a tailored substrate **902** cut to fit the portion **212** of the surface **210** of the additively manufactured structure **200** after tacking the textile substrate **300** to the surface **210**. FIG. 10, in combination with FIG. 1, provides an example of the method **1000** for reinforcing an additively manufactured structure **200**. FIG. 11, in combination with FIG. 1, provides an example of the method **1100** for reinforcing an additively manufactured structure **200**. FIG. 12, in combination with FIG. 1, provides an example of the method **1200** for reinforcing an additively manufactured structure **200**. FIG. 13, in combination with FIG. 1, provides an example of the method **1300** for reinforcing an additively manufactured structure.

[0057] FIG. 14, in combination with FIG. 1, provides an example of the method **1400** for reinforcing an additively manufactured structure **200**. FIG. 15, in combination with FIG. 1, provides an example of the method **1500** for reinforcing an additively manufactured structure **200**. FIG. 16, in combination with FIG. 1, provides an example of the method **1600** for reinforcing an additively manufactured structure. FIGS. 17A-C provides examples of first, second and third textile substrates **300**, **702**, **704**. FIG. 18 shows an example of the application of electric current **402** to conductive fibers **302**, **1702**, **1704** of first, second and third textile substrates **300**, **702**, **704** to form first, second and third thermoplastic composites **404**, **1802**, **1804** on portions **212**, **214**, **216** of the surface **210** of the additively manufactured structure **200**. FIG. 19, in combination with FIGS. 1 and 16, provides an example of the method **1900** for reinforcing an additively manufactured structure **200**.

[0058] FIG. 20, in combination with FIG. 1, provides an example of the method **2000** for reinforcing an additively manufactured structure **200**. FIG. 21 provides another example of a textile substrate **300**. FIG. 22, in combination with FIGS. 1 and 20, provides an example of the method **2200** for reinforcing an additively manufactured structure **200**. FIG. 23 provides examples of the adding **2002** of the one or more conductive fibers **302** in the method **2000** for reinforcing an additively manufactured structure **200**. FIG. 24 provides an example of a carrier sheet **2400** marked for cutting one or more carrier substrates **2102**, **2402**, **2404**.

[0059] FIG. 25, in combination with FIGS. 1 and 20, provides an example of the method **2500** for reinforcing an additively manufactured structure **200**. FIGS. 26A-C provides additional examples of first, second and third textile substrates **300**, **2604**, **2610**. FIG. 27 shows another example of application of electric current **402** to conductive fibers **302**, **2602**, **2608** of first, second and third textile substrates **300**, **2604**, **2610** to form first, second and third thermoplastic composites **404**, **2702**, **2704** on portions **212**, **214**, **216** of the surface **210** of the additively manufactured structure **200**. FIG. 28, in combination with FIGS. 1, 20 and 25, provides an example of the method **2800** for reinforcing an additively manufactured structure **200**. FIG. 39 provides an example of a system **3900** for reinforcing an additively manufactured structure **200**.

[0060] With reference again to FIGS. 1, 2A-B, 3 and 4, in one or more examples, a method **100** (see FIG. 1) for reinforcing an additively manufactured structure **200** includes draping **102** a textile substrate **300** including one or more conductive fibers **302** over at least a portion **212** of a surface **210** of the additively manufactured structure **200**. The additively manufactured structure **200** includes a thermoplastic matrix material **202**. At **104**, an electric current **402** is applied to at least one conductive fiber **302** of the one or more conductive fibers **302** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the textile substrate **300** with the thermoplastic matrix material **202** to form a thermoplastic composite **404** on the

surface **210**.

[0061] In another example of the method **100**, the applying **104** of the electric current **402** to the at least one conductive fiber **302** on the surface **210** of the additively manufactured structure **200** forms a fiber-reinforced additively manufactured structure **406**. In yet another example of the method **100**, the additively manufactured structure **200** includes an outer portion **204** and at least the outer portion **204** includes the thermoplastic matrix material **202**. In still another example of the method **100**, the additively manufactured structure **200** includes an inner portion **206** and the inner portion **206** includes another material **208** different from the thermoplastic matrix material **202**. In a further example, the another material **208** includes a polymer material, a ceramic material, a metal material or any other suitable material in any suitable combination.

[0062] In still yet another example of the method **100**, the thermoplastic matrix material **202** includes an acrylic matrix material, a polycarbonate matrix material, a polyphenylene sulfide matrix material, a polypropylene matrix material, a polystyrene matrix material, a polysulphone matrix material, a polyvinylchloride matrix material or any other suitable thermoplastic matrix material in any suitable combination. In another example of the method **100**, the additively manufactured structure **200** was fabricated using a three-dimensional printer. In yet another example of the method **100**, the additively manufactured structure **200** was fabricated using fused deposition modelling.

[0063] In still another example of the method **100**, the textile substrate **300** includes a woven fabric ply, a braided fabric ply, a knitted fabric ply, a woven film substrate, a woven mesh substrate or any other textile substrate in any suitable combination. In still yet another example of the method **100**, the textile substrate **300** includes natural reinforcement fibers, synthetic reinforcement fibers, non-conductive reinforcement fibers, glass reinforcement fibers, nylon reinforcement fibers, conductive reinforcement fibers, carbon reinforcement fibers, aramid reinforcement fibers, metal reinforcement fibers, copper reinforcement fibers, aluminum reinforcement fibers or any other suitable type of reinforcement fibers in any suitable combination.

[0064] In another example of the method **100**, the textile substrate **300** includes a plurality of the conductive fibers **302** arranged in a first direction **304** and a plurality of non-conductive fibers **306** arranged in a second direction **308** that crosses the first direction **304**. In a further example, the plurality of the conductive fibers **302** includes carbon reinforcement fibers, aramid reinforcement fibers, metal reinforcement fibers, copper reinforcement fibers, aluminum reinforcement fibers or any other suitable type of conductive reinforcement fibers in any suitable combination. In another further example, the plurality of non-conductive fibers **306** include glass reinforcement fibers, nylon reinforcement fibers or any other suitable non-conductive reinforcement fibers in any suitable combination. In yet another example of the method **100**, the textile substrate **300** includes reinforcement fibers pre-impregnated with a resin matrix material. In still another example of the method **100**, the textile substrate **300** is sized to cover the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** over which the textile substrate **300** is draped.

[0065] With reference again to FIGS. **1**, **2**, **3**, **5** and **7**, in one or more examples, a method **500** (see FIG. **5**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The method **500** begins at **502** where the textile substrate **300** is cut from a textile fabric sheet **700** such that the textile substrate **300** is tailored to fit the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** over which the textile substrate **300** is draped. The method **500** continues from **502** to **102** of FIG. **1**.

[0066] In another example of the method **100** of FIG. **1**, the one or more conductive fibers **302** are not in electrical communication with each other within the textile substrate **300**. In yet another example of the method **100**, one or more conductive fibers **302** include one or more carbon fibers, one or more aramid fibers, one or more metal fibers, one or more copper fibers, one or more aluminum fibers or any suitable type of conductive fibers in any suitable combination. In still

another example of the method **100**, at least one of the one or more conductive fibers **302** includes an insulation coating.

[0067] With reference again to FIGS. **1**, **2A-B**, **3** and **6**, in one or more examples, a method **600** (see FIG. **6**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The method **600** continues from **102** of FIGS. **1** to **602** where the textile substrate **300** is tacked to the surface **210** of the additively manufactured structure **200** in conjunction with the draping **102** of the textile substrate **300**. The method **600** continues from **602** to **104** of FIG. **1**.

[0068] With reference again to FIGS. **1**, **2A-B**, **8** and **9**, in another example of the method **100**, the draping **102** of the textile substrate **300** includes wrapping **802** (see FIG. **8**) the textile substrate **300** around the surface **210** of the additively manufactured structure **200**. At **804**, the textile substrate **300** is tacked to the surface **210** of the additively manufactured structure **200** in conjunction with the wrapping **802**. In a further example of the method **800**, the draping **102** of the textile substrate **300** also includes cutting **806** off overlapping portions of the textile substrate **300** in conjunction with the wrapping **802** and tacking **804** of the textile substrate **300** to form a tailored substrate **902** that fits the surface **210**.

[0069] With reference again to FIGS. **1**, **2A-B**, **3**, **4** and **10**, in one or more examples, a method **1000** (see FIG. **10**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The method **1000** continues from **104** of FIGS. **1** to **1002** where the electric current **402** is applied to additional conductive fibers **302** of the one or more conductive fibers **302** to further heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for further consolidation of the textile substrate **300** with the thermoplastic matrix material **202** of the additively manufactured structure **200** to further form the thermoplastic composite **404** on the surface **210**. At **1004**, the electric current **402** is removed from the at least one conductive fiber **302** and the additional conductive fibers **302**.

[0070] With reference again to FIGS. **1**, **2A-B**, **3**, **4** and **11**, in one or more examples, a method **1100** (see FIG. **11**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The consolidation of the textile substrate **300** with the thermoplastic matrix material **202** is based at least in part on the electric current **402** being applied to the at least one conductive fiber **302** for a predetermined time. The method **1100** continues from **104** of FIGS. **1** to **1102** to monitor a time the electric current **402** was applied to the at least one conductive fiber **302**. At **1104**, the electric current **402** is removed from the at least one conductive fiber **302** after the time the electric current **402** was applied exceeds the predetermined time.

[0071] With reference again to FIGS. **1**, **2A-B**, **3**, **4** and **12**, in one or more examples, a method **1200** (see FIG. **12**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The consolidation of the textile substrate **300** with the thermoplastic matrix material **202** is based at least in part on the electric current **402** being applied to the at least one conductive fiber **302** for a predetermined time. The method **1200** continues from **104** of FIGS. **1** to **1202** to monitor a temperature of the thermoplastic matrix material **202** proximate the at least one conductive fiber **302**. At **1204**, the electric current **402** is removed from the at least one conductive fiber **302** after the temperature exceeds the predetermined temperature.

[0072] With reference again to FIGS. **1**, **2A-B**, **3**, **13** and **39**, in one or more examples, a method **1300** (see FIG. **13**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The consolidation of the textile substrate **300** with the thermoplastic matrix material **202** is based at least in part on applying a compressive force to the textile substrate **300**. The method **1100** continues from **104** of FIGS. **1** to **1302** to use a weighted cover **3918** over the textile substrate **300**, a vacuum bagging subsystem **3920** or any other suitable consolidation tool in any suitable combination.

[0073] With reference again to FIGS. **1**, **4** and **14**, in one or more examples, a method **1400** (see FIG. **14**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The method **1400** continues from **104** of FIGS. **1** to **1402** where the electric current **402** is

removed from the at least one conductive fiber **302**.

[0074] With reference again to FIGS. **1**, **2A-B**, **4** and **15**, in one or more examples, a method **1500** (see FIG. **15**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The method **1500** continues from **104** of FIGS. **1** to **1502** where the thermoplastic composite **404** on the surface **210** of the additively manufactured structure **200** is cured to form a fiber-reinforced additively manufactured structure **406**. In another example of the method **1500**, the curing **1502** of the thermoplastic composite **404** is performed using a local ambient curing process, a local heating process, an oven curing process, an ultraviolet light curing process, a visible light curing process, a direct electric heating process, a microwave heating process, an induction heating process, a radio frequency heating process or any other suitable curing process in any suitable combination. The method **1500** also continues from **104** of FIGS. **1** to **1504** where the thermoplastic composite **404** is trimmed to remove waste materials resulting from consolidation of the textile substrate **300** with the thermoplastic matrix material **202**.

[0075] With reference again to FIGS. **1**, **2A-B**, **3**, **4** and **16-18**, in one or more examples, a method **1600** (see FIG. **16**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The textile substrate **300** is cut from a first textile fabric sheet **700** such that the textile substrate **300** is tailored to fit a first portion **212** of the surface **210** of the additively manufactured structure **200**. The method **1600** continues from **104** of FIGS. **1** to **1602** where a second textile substrate **702** including one or more second conductive fibers **1702** is cut from the first textile fabric sheet **700** or a second textile fabric sheet **700'** such that the second textile substrate **702** is tailored to fit a second portion **214** of the surface **210**. At **1604**, the second textile substrate **702** is draped over the second portion **214** of the surface **210**. At **1606**, the electric current **402** is applied to at least one second conductive fiber **1702** of the one or more second conductive fibers **1702** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the second textile substrate **702** with the thermoplastic matrix material **202** to form a second thermoplastic composite **1802** on the surface **210**.

[0076] With reference again to FIGS. **1**, **2A-B**, **7**, **16**, **18** and **19**, in one or more examples, a method **1900** (see FIG. **19**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1** and the method **1600** of FIG. **16**. The method **1900** continues from **1606** of FIGS. **16** to **1902** where a third textile substrate **704** including one or more third conductive fibers **1704** is cut from the first textile fabric sheet **700** or a third textile fabric sheet **700''** such that the third textile substrate **704** is tailored to fit a third portion **216** of the surface **210** of the additively manufactured structure **200**. At **1904**, the third textile substrate **704** is draped over the third portion **216** of the surface **210**. At **1906**, the electric current **402** is applied to at least one third conductive fiber **1704** of the one or more third conductive fibers **1704** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the third textile substrate **704** with the thermoplastic matrix material **202** to form a third thermoplastic composite **1804** on the surface **210**.

[0077] With reference again to FIGS. **1**, **2A-B**, **3**, **20** and **21**, in one or more examples, a method **2000** (see FIG. **20**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**. The method **2000** begins at **2002** where the one or more conductive fibers **302** are added to a carrier substrate **2102** to form the textile substrate **300**. The one or more conductive fibers **302** forming one or more paths **2104** distributed about the textile substrate **300**. The method **2000** continues from **2002** to **102** of FIG. **1**.

[0078] In another example of the method **2000**, the one or more conductive fibers **302** are added to the carrier substrate **2102** using tailored fiber placement. In yet another example of the method **2000**, the one or more conductive fibers **302** include carbon reinforcement fibers, aramid reinforcement fibers, metal reinforcement fibers, copper reinforcement fibers, aluminum reinforcement fibers or any other suitable conductive reinforcement fiber in any suitable combination.

[0079] In still another example of the method **2000**, the carrier substrate **2102** includes a woven fabric ply, a braided fabric ply, a knitted fabric ply, a woven film substrate, a woven mesh substrate, a non-woven fabric substrate, a plastic film substrate, a laminate film substrate, a mesh screen substrate or any other suitable carrier substrate in any suitable combination. In still yet another example of the method **2000**, the carrier substrate **2102** includes non-conductive reinforcement fibers, glass reinforcement fibers, nylon reinforcement fibers or any other suitable reinforcement fiber in any suitable combination. In another example of the method **2000**, the carrier substrate **2102** is sized to cover the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** over which the textile substrate **300** is draped.

[0080] With reference again to FIGS. **1**, **2A-B**, **3**, **20**, **22**, **23** and **24**, in one or more examples, a method **2200** (see FIG. **22**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1** and the method **2000** of FIG. **20**. The method **2200** begins at **2202** where the carrier substrate **2102** is cut from a carrier sheet **2400** such that the carrier substrate **2102** is tailored to fit the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** over which the textile substrate **300** is draped. The method **2200** continues from **2202** to **2002** of FIG. **20**.

[0081] In yet another example of the method **2000**, the adding **2002** of the one or more conductive fibers **302** includes embroidering **2302** at least one of the one or more conductive fibers **302** to the carrier substrate **2102**. In still another example of the method **2000**, the adding **2002** of the one or more conductive fibers **302** includes sewing **2304** at least one of the one or more conductive fibers **302** to the carrier substrate **2102**.

[0082] With reference again to FIGS. **1**, **2A-B**, **4**, **20**, **21** and **24-27**, in one or more examples, a method **2500** (see FIG. **25**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1** and the method **2000** of FIG. **20**. The carrier substrate **2102** is cut from a first carrier sheet **2400** such that the carrier substrate **2102** is tailored to fit a first portion **212** of the surface **210** of the additively manufactured structure **200**. The method **2500** continues from **2002** of FIGS. **20** to **2502** where a second carrier substrate **2402** is cut from the first carrier sheet **2400** or a second carrier sheet **2400'** such that the second carrier substrate **2402** is tailored to fit a second portion **214** of the surface **210**. At **2504**, one or more second conductive fibers **2602** are added to the second carrier substrate **2402** to form a second textile substrate **2604**. The one or more second conductive fibers **2602** forming one or more second paths **2606** distributed about the second textile substrate **2604**. At **2506**, the second textile substrate **2604** is draped over the second portion **214** of the surface **210**. At **2508**, the electric current **402** is applied to at least one second conductive fiber **2602** of the one or more second conductive fibers **2602** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the second textile substrate **2604** with the thermoplastic matrix material **202** to form a second thermoplastic composite **2702** on the surface **210**.

[0083] With reference again to FIGS. **1**, **2A-B**, **4**, **20**, **24** and **26-28**, in one or more examples, a method **2800** (see FIG. **28**) for reinforcing an additively manufactured structure **200** includes the method **100** of FIG. **1**, the method **2000** of FIG. **20** and the method **2500** of FIG. **25**. The method **2800** continues from **2508** of FIGS. **25** to **2802** where a third carrier substrate **2404** is cut from the first carrier sheet **2400** or a third carrier sheet **2400''** such that the third carrier substrate **2404** is tailored to fit a third portion **216** of the surface **210** of the additively manufactured structure **200**. At **2804**, one or more third conductive fibers **2608** are added to the third carrier substrate **2404** to form a third textile substrate **2610**. The one or more third conductive fibers **2608** forming one or more third paths **2612** distributed about the third textile substrate **2610**. At **2806**, the third textile substrate **2610** is draped over the third portion **216** of the surface **210**. At **2808**, the electric current **402** is applied to at least one third conductive fiber **2608** of the one or more third conductive fibers **2608** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the third textile substrate **2610** with the thermoplastic matrix material **202** to form

a third thermoplastic composite **2704** on the surface **210**.

[0084] Referring generally to FIGS. **2A-B** and **29-38**, by way of examples, the present disclosure is directed to a method **2900**, **3600**, **3700**, **3800** for reinforcing an additively manufactured structure **200**. FIG. **2A** provides an example of an additively manufactured structure **200**. FIG. **2B** provides a cross-sectional view of the additively manufactured structure **200**. FIG. **29** provides an example of the method **2900** for reinforcing an additively manufactured structure **200**. FIG. **30** provides an example of a textile substrate **3000** marked for cutting a tailored fabric substrate **3100**. FIG. **31** provides an example of the tailored fabric substrate **3100**. FIG. **32** shows an example of the application of electric current to first, second and third conductive fibers **3002**, **3010**, **3016** of the tailored fabric substrate **3100** to form first, second and third thermoplastic composites **3204**, **3208**, **3210** on portions **212**, **214**, **216** of the surface **210** of the additively manufactured structure **200**.

[0085] FIG. **33** provides examples of the adding **2902** of the first conductive fiber **3002** in the method **2900** for reinforcing an additively manufactured structure **200**. FIG. **34** provides an example of the wrapping **2906** of the tailored fabric substrate **3100** in the method **2900** for reinforcing an additively manufactured structure **200**. FIG. **35** provides an example of a tailored substrate **3502** cut to fit a portion **212**, **214**, **216** of the surface **210** of the additively manufactured structure **200** after tacking **3402** the tailored fabric substrate **3502** to the surface **210**. FIG. **36**, in combination with FIG. **29**, provides an example of the method **3600** for reinforcing an additively manufactured structure **200**. FIG. **37**, in combination with FIG. **29**, provides an example of the method **3700** for reinforcing an additively manufactured structure **200**. FIG. **38**, in combination with FIGS. **29** and **37**, provides an example of the method **3800** for reinforcing an additively manufactured structure **200**.

[0086] With reference again to FIGS. **2A-B** and **29-35**, in one or more examples, a method **2900** (see FIG. **29**) for reinforcing an additively manufactured structure **200** includes adding **2902** a first conductive fiber **3002** to a fiber reinforcement fabric substrate **3004** to form a textile substrate **3000**. The first conductive fiber **3002** forming a first path **3006** distributed about at least a first area **3008** of the textile substrate **3000**. At **2904**, the textile substrate **3000** is cut to form a tailored fabric substrate **3100** that fits a surface **210** of the additively manufactured structure **200**. The additively manufactured structure **200** includes a thermoplastic matrix material **202**. At **2906**, the tailored fabric substrate **3100** is wrapped around the surface **210**. At **2908**, an electric current **3202** is applied to the first conductive fiber **3002** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the tailored fabric substrate **3100** with the thermoplastic matrix material **202** to form a first thermoplastic composite **3204** on the surface **210** and thereby form a fiber-reinforced additively manufactured structure **3206**.

[0087] In another example of the method **2900**, the additively manufactured structure **200** includes an outer portion **204** and an inner portion **206**. The outer portion **204** includes the thermoplastic matrix material **202**. The inner portion **206** includes another material **208** different from the thermoplastic matrix material **202**. In a further example, the another material **208** includes a polymer material, a ceramic material, a metal material or any other suitable material in any suitable combination. In yet another example of the method **2900**, the thermoplastic matrix material **202** includes an acrylic matrix material, a polycarbonate matrix material, a polyphenylene sulfide matrix material, a polypropylene matrix material, a polystyrene matrix material, a polysulphone matrix material, a polyvinylchloride matrix material or any other suitable thermoplastic matrix material in any suitable combination.

[0088] In still another example of the method **2900**, the additively manufactured structure **200** was fabricated using a three-dimensional printer. In still yet another example of the method **2900**, the additively manufactured structure **200** was fabricated using fused deposition modelling. In another example of the method **2900**, the first conductive fiber **3002** is added to the fiber reinforcement fabric substrate **3004** using tailored fiber placement.

[0089] In yet another example of the method **2900**, the fiber reinforcement fabric substrate **3004**

includes a woven fabric ply, a woven film substrate, a woven mesh substrate or any other suitable fiber reinforcement fabric substrate in any suitable combination. In still another example of the method **2900**, the fiber reinforcement fabric substrate **3004** includes non-conductive reinforcement fibers, glass reinforcement fibers, nylon reinforcement fibers, conductive reinforcement fibers, carbon reinforcement fibers, metal reinforcement fibers, copper reinforcement fibers, aluminum reinforcement fibers or any suitable reinforcement fibers in any suitable combination. In still yet another example of the method **2900**, the fiber reinforcement fabric substrate **3004** includes reinforcement fibers pre-impregnated with a resin matrix material. In another example of the method **2900**, the fiber reinforcement fabric substrate **3004** is sized to cover the surface **210** of the additively manufactured structure **200**.

[0090] In yet another example of the method **2900**, the first conductive fiber **3002** includes a carbon fiber, an aramid fiber, a metal fiber, a copper fiber, an aluminum fiber or any other suitable conductive fiber in any suitable combination. In still another example of the method **2900**, the first conductive fiber **3002** includes an insulation coating.

[0091] In still yet another example of the method **2900**, the adding **2902** of the first conductive fiber **3002** includes embroidering **3302** the first conductive fiber **3002** to a fiber reinforcement fabric substrate **3004**. In another example of the method **2900**, the adding **2902** of the first conductive fiber **3002** includes sewing **3304** the first conductive fiber **3002** to the fiber reinforcement fabric substrate **3004**.

[0092] In yet another example of the method **2900**, the wrapping **2906** of the tailored fabric substrate **3100** includes tacking **3402** the tailored fabric substrate **3100** to the surface **210** of the additively manufactured structure **200** in conjunction with the wrapping **2906**. In still another example of the method **2900**, the wrapping **2906** of the tailored fabric substrate **3100** also includes cutting **3404** off overlapping portions of the tailored fabric substrate **3100** during the wrapping **2906** and tacking **3402** to form a tailored substrate **3502** that fits the surface **210**.

[0093] With reference again to FIGS. 2A-B, 29-32 and 36, in one or more examples, a method **3600** (see FIG. 36) for reinforcing an additively manufactured structure **200** includes the method **2900** of FIG. 29. The method **3600** continues from **2908** of FIGS. 29 to **3602** where the electric current **3202** is removed from the first conductive fiber **3002**. The method **3600** also continues from **2908** of FIGS. 29 to **3604** where the first thermoplastic composite **3204** is trimmed to remove waste materials resulting from consolidation of the tailored fabric substrate **3100** with the thermoplastic matrix material **202**. Additionally, the method **3600** continues from **2908** of FIGS. 29 to **3606** where the first thermoplastic composite **3204** on the surface **210** of the additively manufactured structure **200** is cured to form the fiber-reinforced additively manufactured structure **3206**. In a further example, the curing **3606** of the first thermoplastic composite **3204** is performed using a local ambient curing process, a local heating process, an oven curing process, an ultraviolet light curing process, a visible light curing process, a direct electric heating process, a microwave heating process, an induction heating process, a radio frequency heating process or any other suitable curing process in any suitable combination.

[0094] With reference again to FIGS. 2A-B, 29-32 and 37, in one or more examples, a method **3700** (see FIG. 37) for reinforcing an additively manufactured structure **200** includes the method **2900** of FIG. 29. The method **3700** continues from **2902** of FIGS. 29 to **3702** where a second conductive fiber **3010** is added to the textile substrate **3000**. The second conductive fiber **3010** forming a second path **3012** distributed about a second area **3014** of the textile substrate **3000**. After **3702**, the method **3700** continues to **2904** of FIG. 29. Additionally, the method **3700** continues from **2908** of FIGS. 29 to **3704** where the electric current **3202** is applied to heat the second conductive fiber **3010** for further consolidation of the tailored fabric substrate **3100** with the thermoplastic matrix material **202** of the additively manufactured structure **200** to form a second thermoplastic composite **3208** on the surface **210** and thereby further form the fiber-reinforced additively manufactured structure **3206**.

[0095] With reference again to FIGS. 2A-B, 29-32, 37 and 38, in one or more examples, a method **3800** (see FIG. 38) for reinforcing an additively manufactured structure **200** includes the method **2900** of FIG. 29 and the method **3700** of FIG. 37. The method **3800** continues from **3702** of FIGS. 37 to **3802** where a third conductive fiber **3016** is added to the textile substrate **3000**. The third conductive fiber **3016** forming a third path **3018** distributed about a third area **3020** of the textile substrate **3000**. After **3802**, the method **3800** continues to **2904** of FIG. 29. Additionally, the method **3800** continues from **2908** of FIGS. 29 to **3804** where the electric current **3202** is applied to heat the third conductive fiber **3016** for further consolidation of the tailored fabric substrate **3100** with the thermoplastic matrix material **202** of the additively manufactured structure **200** to form a third thermoplastic composite **3210** on the surface **210** and thereby further form the fiber-reinforced additively manufactured structure **3206**.

[0096] Referring generally to FIGS. 1, 2A-B, 3, 4, 17A-C, 18, 21, 24, 26, 27, 30, 39 and 40, by way of examples, the present disclosure is directed to a system **3900**, **4000** for reinforcing an additively manufactured structure **200**. FIG. 1 provides an example of a method **100** for reinforcing an additively manufactured structure **200**. FIG. 2A provides an example of an additively manufactured structure **200**. FIG. 2B provides a cross-sectional view of the additively manufactured structure **200**. FIG. 3 provides an example of a textile substrate **300**. FIG. 4 shows an example of the application of electric current **402** to conductive fibers **302** of the textile substrate **300** to form a thermoplastic composite **404** on a portion **212** of the surface **210** of the additively manufactured structure **200**. FIGS. 17A-C provides examples of first, second and third textile substrates **300**, **702**, **704**. FIG. 18 shows an example of the application of electric current **402** to conductive fibers **302**, **1702**, **1704** of first, second and third textile substrates **300**, **702**, **704** to form first, second and third thermoplastic composites **404**, **1802**, **1804** on portions **212**, **214**, **216** of the surface **210** of the additively manufactured structure **200**.

[0097] FIG. 21 provides another example of a textile substrate **300**. FIG. 24 provides an example of a carrier sheet **2400** marked for cutting one or more carrier substrates **2102**, **2402**, **2404**. FIGS. 26A-C provides additional examples of first, second and third textile substrates **300**, **2604**, **2610**. FIG. 27 shows another example of application of electric current **402** to conductive fibers **302**, **2602**, **2608** of first, second and third textile substrates **300**, **2604**, **2610** to form first, second and third thermoplastic composites **404**, **2702**, **2704** on portions **212**, **214**, **216** of the surface **210** of the additively manufactured structure **200**. FIG. 39 provides an example of the system **3900** for reinforcing an additively manufactured structure **200**. FIG. 40 provides an example of the system **4000** for reinforcing an additively manufactured structure **200**.

[0098] With reference again to FIGS. 1, 2A-B, 3, 4, 7, 9, 17, 18, 30 and 39, in one or more examples, a system **3900** (see FIG. 39) for reinforcing an additively manufactured structure **200** includes a fabrication tool set **3902**, an electrical power source **3904** and a control device **3906**. The control device **3906** in operative communication with the electrical power source **3904** to selectively apply and remove electric current **402** from one or more conductive fibers **302** of a textile substrate **300**. The textile substrate **300** is configured for draping over at least a portion **212** of a surface **210** of the additively manufactured structure **200**. The additively manufactured structure **200** includes a thermoplastic matrix material **202**. The control device **3906** and the electrical power source **3904** are configured to apply the electric current **402** to at least one conductive fiber **302** of the one or more conductive fibers **302** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the textile substrate **300** with the thermoplastic matrix material **202** to form a thermoplastic composite **404** on the surface **210**.

[0099] In another example of the system **3900**, the additively manufactured structure **200** was fabricated using a three-dimensional printer. In yet another example of the system **3900**, the additively manufactured structure **200** was fabricated using fused deposition modelling. In still another example of the system **3900**, the textile substrate **300** is sized to cover the surface **210** of

the additively manufactured structure **200** or at least the portion **212** of the surface **210** of the additively manufactured structure **200**.

[0100] In still yet another example of the system **3900**, the fabrication tool set **3902** includes a cutting tool **3908** configured to cut the fiber reinforcement fabric substrate **3004** from a fiber reinforcement fabric sheet **900** such that the fiber reinforcement fabric substrate **3004** is tailored to fit the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** of the additively manufactured structure **200** over which the textile substrate **300** is draped. In another example of the system **3900**, the fabrication tool set **3902** includes a tacking tool **3910** configured to tack the textile substrate **300** to the surface **210** of the additively manufactured structure **200** in conjunction with the draping **102** of the of the textile substrate **300**.

[0101] In yet another example of the system **3900**, the textile substrate **300** is configured for being wrapped around the surface **210** of the additively manufactured structure **200**. In this example, the fabrication tool set **3902** includes a tacking tool **3910** configured to tack the textile substrate **300** to the surface **210** in conjunction with the wrapping of the textile substrate **300**. In a further example, the fabrication tool set **3902** also includes a cutting tool **3908** configured to cut off overlapping portions of the textile substrate **300** in conjunction with the wrapping **802** and tacking **804** of the textile substrate **300** to form a tailored substrate **902** that fits the surface **210**.

[0102] In still another example of the system **3900**, the control device **3906** and the electrical power source **3904** are configured to apply the electric current **402** to additional conductive fibers **302** of the one or more conductive fibers **302** to further heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for further consolidation of the textile substrate **300** with the thermoplastic matrix material **202** of the additively manufactured structure **200** to further form the thermoplastic composite **404**. In a further example, the control device **3906** and the electrical power source **3904** are configured to remove the electric current **402** from the additional conductive fibers **302**.

[0103] In still yet another example of the system **3900**, the consolidation of the textile substrate **300** with the thermoplastic matrix material **202** is based at least in part on the electric current **402** being applied to the at least one conductive fiber **302** for a predetermined time. In this example, the control device **3906** includes a timing device **3912** configured to monitor a time the electric current **402** was applied to the at least one conductive fiber **302**. The control device **3906** and the electrical power source **3904** are configured to remove the electric current **402** from the at least one conductive fiber **302** after the timing device **3912** indicates the time the electric current **402** was applied exceeds the predetermined time.

[0104] In another example of the system **3900**, the consolidation of the textile substrate **300** with the thermoplastic matrix material **202** is based at least in part on the thermoplastic matrix material **202** being heated to a predetermined temperature. In this example, the system also includes a temperature sensor **3914** in operative communication with the control device **3906** and disposed on the additively manufactured structure **200** proximate the at least one conductive fiber **302**. The temperature sensor **3914** is configured to monitor a temperature of the thermoplastic matrix material **202** proximate the at least one conductive fiber **302**. The control device **3906** and the electrical power source **3904** are configured to remove the electric current **402** from the at least one conductive fiber **302** after the temperature sensor **3914** indicates the temperature exceeds the predetermined temperature.

[0105] In yet another example of the system **3900**, the consolidation of the textile substrate **300** with the thermoplastic matrix material **202** is based at least in part on applying a compressive force to the textile substrate **300**. In this example, the fabrication tool set **3902** includes a consolidation tool **3916** configured to apply a compressive force to the textile substrate **300** for consolidation of the textile substrate **300** with the thermoplastic matrix material **202**. In a further example, the consolidation tool **3916** includes a weighted cover **3918**, a vacuum bagging subsystem **3920** or any other suitable consolidation tool in any suitable combination.

[0106] In still another example of the system **3900**, the control device **3906** and the electrical power source **3904** are configured to remove the electric current **402** from the at least one conductive fiber **302**.

[0107] In still yet another example of the system **3900**, the fabrication tool set **3902** includes a curing subsystem **3922** configured to cure the thermoplastic composite **404** on the surface **210** of the additively manufactured structure **200** to form a fiber-reinforced additively manufactured structure **406**. In a further example, the curing subsystem **3018** includes a local heating device, an oven curing device, an ultraviolet light curing device, a visible light curing device, a direct electric heating device, a microwave heating device, an induction heating device, a radio frequency heating device or any other suitable curing subsystem in any suitable combination. In another example of the system **3900**, the fabrication tool set **3902** includes a trimming tool **3924** configured to trim the thermoplastic composite **404** to remove waste materials.

[0108] In yet another example of the system **3900**, the fabrication tool set **3902** includes a cutting tool **3908** configured to cut the textile substrate **300** from a first textile fabric sheet **700** such that the textile substrate **300** is tailored to fit a first portion **212** of the surface **210** of the additively manufactured structure **200**. The cutting tool **3908** is configured to cut a second textile substrate **702** from the first textile fabric sheet **700** or a second textile fabric sheet **700'** such that the second textile substrate **702** is tailored to fit a second portion **214** of the surface **210**. The second textile substrate **702** is configured for draping over the second portion **214** of the surface **210**. The control device **3906** and the electrical power source **3904** are configured to apply the electric current **402** to at least one second conductive fiber **1702** of the one or more second conductive fibers **1702** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the second textile substrate **702** with the thermoplastic matrix material **202** to form a second thermoplastic composite **1802** on the surface **210**.

[0109] In a further example, the cutting tool **3908** is configured to cut a third textile substrate **704** from the first textile fabric sheet **700** or a third textile fabric sheet **700''** such that the third textile substrate **704** is tailored to fit a third portion **216** of the surface **210**. The third textile substrate **704** is configured for draping over the third portion **216** of the surface **210**. The control device **3906** and the electrical power source **3904** are configured to apply the electric current **402** to at least one third conductive fiber **1704** of the one or more third conductive fibers **1704** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the third textile substrate **704** with the thermoplastic matrix material **202** to form a third thermoplastic composite **1804** on the surface **210**.

[0110] With reference again to FIGS. 2A-B, 3, 21, 24, 26, 27, 39 and 40, in one or more examples, a system **4000** (see FIG. 40) for reinforcing an additively manufactured structure **200** includes the system **3900** of FIG. 39. In the system **4000**, the fabrication tool set **3902** includes a fiber adding tool **4002** configured to add the one or more conductive fibers **302** to a carrier substrate **2102** to form the textile substrate **300**. The one or more conductive fibers **302** forming one or more paths **2104** distributed about the textile substrate **300**.

[0111] In another example of the system **4000**, the one or more conductive fibers **302** are added to the carrier substrate **2102** using tailored fiber placement. In yet another example of the system **4000**, the carrier substrate **2102** is sized to cover the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** over which the textile substrate **300** is draped.

[0112] In still another example of the system **4000**, the fiber adding tool **4002** includes an embroidery machine **4004** configured to embroider at least one of the one or more conductive fibers **302** to the carrier substrate **2102**. In still yet another example of the system **4000**, the fiber adding tool **4002** includes a sewing machine **4006** configured to sew at least one of the one or more conductive fibers **302** to the carrier substrate **2102**.

[0113] In another example of the system **4000**, the fabrication tool set **3902** also includes a cutting

tool **3908** configured to cut the carrier substrate **2102** from a carrier sheet **2400** such that the carrier substrate **2102** is tailored to fit the surface **210** of the additively manufactured structure **200** or at least the portion **212** of the surface **210** over which the textile substrate **300** is draped.

[0114] In a further example, the cutting tool **3908** is configured to cut the carrier substrate **2102** from a first carrier sheet **2400** such that the carrier substrate **2102** is tailored to fit a first portion **212** of the surface **210** of the additively manufactured structure **200**. The cutting tool **3908** is configured to cut a second carrier substrate **2402** from the first carrier sheet **2400** or a second carrier sheet **2400'** such that the second carrier substrate **2402** is tailored to fit a second portion **214** of the surface **210**. The fiber adding tool **4002** is configured to add one or more second conductive fibers **2602** to the second carrier substrate **2402** to form a second textile substrate **2604**. The one or more second conductive fibers **2602** forming one or more second paths **2606** distributed about the second textile substrate **2604**. The second textile substrate **2604** is configured for draping over the second portion **214** of the surface **210**. The control device **3906** and the electrical power source **3904** are configured to apply the electric current **402** to at least one second conductive fiber **2602** of the one or more second conductive fibers **2602** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the second textile substrate **2604** with the thermoplastic matrix material **202** to form a second thermoplastic composite **2702** on the surface **210**.

[0115] In another further example, the cutting tool **3908** is configured to cut a third carrier substrate **2404** from the first carrier sheet **2400** or a third carrier sheet **2400''** such that the third carrier substrate **2404** is tailored to fit a third portion **216** of the surface **210** of the additively manufactured structure **200**. The fiber adding tool **4002** is configured to add one or more third conductive fibers **2608** to the third carrier substrate **2404** to form a third textile substrate **2610**. The one or more third conductive fibers **2608** forming one or more third paths **2612** distributed about the third textile substrate **2610**. The third textile substrate **2610** is configured for draping over the third portion **216** of the surface **210**. The control device **3906** and the electrical power source **3904** are configured to apply the electric current **402** to at least one third conductive fiber **2608** of the one or more third conductive fibers **2608** to heat the thermoplastic matrix material **202** of the additively manufactured structure **200** for consolidation of the third textile substrate **2610** with the thermoplastic matrix material **202** to form a third thermoplastic composite **2704** on the surface **210**.

[0116] Examples of the methods **100**, **500**, **600**, **1000**, **1100**, **1200**, **1300**, **1400**, **1500**, **1600**, **1900**, **2000**, **2200**, **2500**, **2800**, **2900**, **3600**, **3700**, **3800** and systems **3900**, **4000** for reinforcing additively manufactured structures **200** may be related to or used in the context of aircraft manufacturing.

Although an aircraft example is described, the examples and principles disclosed herein may be applied to other products in the aerospace industry and other industries, such as the automotive industry, the space industry, the construction industry and other design and manufacturing industries. Accordingly, in addition to aircraft, the examples and principles disclosed herein may be implemented to produce components and other equipment in various types of vehicles and in the construction of various types of buildings.

[0117] The preceding detailed description refers to the accompanying drawings, which illustrate specific examples described by the present disclosure. Other examples having different structures and operations do not depart from the scope of the present disclosure. Like reference numerals may refer to the same feature, element or component in the different drawings. Throughout the present disclosure, any one of a plurality of items may be referred to individually as the item and a plurality of items may be referred to collectively as the items and may be referred to with like reference numerals. Moreover, as used herein, a feature, element, component or step preceded with the word “a” or “an” should be understood as not excluding a plurality of features, elements, components or steps, unless such exclusion is explicitly recited.

[0118] Illustrative, non-exhaustive examples, which may be, but are not necessarily, claimed, of the subject matter according to the present disclosure are provided above. Reference herein to

“example” means that one or more feature, structure, element, component, characteristic and/or operational step described in connection with the example is included in at least one aspect, embodiment and/or implementation of the subject matter according to the present disclosure. Thus, the phrases “an example,” “another example,” “one or more examples,” and similar language throughout the present disclosure may, but do not necessarily, refer to the same example. Further, the subject matter characterizing any one example may, but does not necessarily, include the subject matter characterizing any other example. Moreover, the subject matter characterizing any one example may be, but is not necessarily, combined with the subject matter characterizing any other example.

[0119] As used herein, a system, apparatus, device, structure, article, element, component or hardware “configured to” perform a specified function is indeed capable of performing the specified function without any alteration, rather than merely having potential to perform the specified function after further modification. In other words, the system, apparatus, device, structure, article, element, component or hardware “configured to” perform a specified function is specifically selected, created, implemented, utilized, programmed and/or designed for the purpose of performing the specified function. As used herein, “configured to” denotes existing characteristics of a system, apparatus, structure, article, element, component or hardware that enable the system, apparatus, structure, article, element, component or hardware to perform the specified function without further modification. For purposes of this disclosure, a system, apparatus, device, structure, article, element, component or hardware described as being “configured to” perform a particular function may additionally or alternatively be described as being “adapted to” and/or as being “operative to” perform that function.

[0120] Unless otherwise indicated, the terms “first,” “second,” “third,” etc. are used herein merely as labels and are not intended to impose ordinal, positional or hierarchical requirements on the items to which these terms refer. Moreover, reference to, e.g., a “second” item does not require or preclude the existence of, e.g., a “first” or lower-numbered item and/or, e.g., a “third” or higher-numbered item.

[0121] As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items may be used and only one of each item in the list may be needed. For example, “at least one of item A, item B and item C” may include, without limitation, item A or item A and item B. This example also may include item A, item B and item C or item B and item C. In other examples, “at least one of” may be, for example, without limitation, two of item A, one of item B and ten of item C; four of item B and seven of item C; and other suitable combinations. As used herein, the term “and/or” and the “/” symbol includes any and all combinations of one or more of the associated listed items.

[0122] As used herein, the terms “coupled,” “coupling,” and similar terms refer to two or more elements that are joined, linked, fastened, attached, connected, put in communication or otherwise associated (e.g., mechanically, electrically, fluidly, optically, electromagnetically) with one another. In various examples, the elements may be associated directly or indirectly. As an example, element A may be directly associated with element B. As another example, element A may be indirectly associated with element B, for example, via another element C. It will be understood that not all associations among the various disclosed elements are necessarily represented. Accordingly, couplings other than those depicted in the figures may also exist.

[0123] As used herein, the term “approximately” refers to or represents a condition that is close to, but not exactly, the stated condition that still performs the desired function or achieves the desired result. As an example, the term “approximately” refers to a condition that is within an acceptable predetermined tolerance or accuracy, such as to a condition that is within 10% of the stated condition. However, the term “approximately” does not exclude a condition that is exactly the stated condition. As used herein, the term “substantially” refers to a condition that is essentially the stated condition that performs the desired function or achieves the desired result.

[0124] In FIGS. **1, 5, 6, 8, 10-16, 19, 20, 22, 23, 25, 28, 29, 33, 34,** and **36-38**, referred to above, the blocks may represent operations, steps and/or portions thereof and lines connecting the various blocks do not imply any particular order or dependency of the operations or portions thereof. It will be understood that not all dependencies among the various disclosed operations are necessarily represented. FIGS. **1, 5, 6, 8, 10-16, 19, 20, 22, 23, 25, 28, 29, 33, 34,** and **36-38** and the accompanying disclosure describing the operations of the disclosed methods set forth herein should not be interpreted as necessarily determining a sequence in which the operations are to be performed. Rather, although one illustrative order is indicated, it is to be understood that the sequence of the operations may be modified when appropriate. Accordingly, modifications, additions and/or omissions may be made to the operations illustrated and certain operations may be performed in a different order or simultaneously. Additionally, those skilled in the art will appreciate that not all operations described need be performed.

[0125] In FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40**, referred to above, may represent functional elements, features or components thereof and do not necessarily imply any particular structure. Accordingly, modifications, additions and/or omissions may be made to the illustrated structure. Additionally, those skilled in the art will appreciate that not all elements, features and/or components described and illustrated in FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40**, referred to above, need be included in every example and not all elements, features and/or components described herein are necessarily depicted in each illustrative example. Accordingly, some of the elements, features and/or components described and illustrated in FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40** may be combined in various ways without the need to include other features described and illustrated in FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40**, other drawing figures and/or the accompanying disclosure, even though such combination or combinations are not explicitly illustrated herein. Similarly, additional features not limited to the examples presented, may be combined with some or all the features shown and described herein. Unless otherwise explicitly stated, the schematic illustrations of the examples depicted in FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40**, referred to above, are not meant to imply structural limitations with respect to the illustrative example. Rather, although one illustrative structure is indicated, it is to be understood that the structure may be modified when appropriate. Accordingly, modifications, additions and/or omissions may be made to the illustrated structure. Furthermore, elements, features and/or components that serve a similar, or at least substantially similar, purpose are labeled with like numbers in each of FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40** and such elements, features and/or components may not be discussed in detail herein with reference to each of FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40**. Similarly, all elements, features and/or components may not be labeled in each of FIGS. **2A-B, 3, 4, 7, 9, 17A-C, 18, 21, 24, 26A-C, 27, 30-32, 35, 39** and **40**, but reference numerals associated therewith may be utilized herein for consistency.

[0126] Further, references throughout the present specification to features, advantages, or similar language used herein do not imply that all the features and advantages that may be realized with the examples disclosed herein should be, or are in, any single example. Rather, language referring to the features and advantages is understood to mean that a specific feature, advantage or characteristic described in connection with an example is included in at least one example. Thus, discussion of features, advantages and similar language used throughout the present disclosure may, but does not necessarily, refer to the same example.

[0127] Examples of the subject matter disclosed herein may be described in the context of aircraft manufacturing and service method **4100** as shown in FIG. **41** and aircraft **4200** as shown in FIG. **42**. In one or more examples, the disclosed methods and systems for associating test data for a part under test with an end item coordinate system may be used in aircraft manufacturing. During pre-production, the service method **4100** may include specification and design (block **4102**) of aircraft

4200 and material procurement (block **4104**). During production, component and subassembly manufacturing (block **4106**) and system integration (block **4108**) of aircraft **4200** may take place. Thereafter, aircraft **4200** may go through certification and delivery (block **1410**) to be placed in service (block **4112**). While in service, aircraft **4200** may be scheduled for routine maintenance and service (block **4114**). Routine maintenance and service may include modification, reconfiguration, refurbishment, etc. of one or more systems of aircraft **4200**.

[0128] Each of the processes of the service method **4100** may be performed or carried out by a system integrator, a third party and/or an operator (e.g., a customer). For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors and suppliers; and an operator may be an airline, leasing company, military entity, service organization and so on.

[0129] As shown in FIG. **42**, aircraft **4200** produced by the service method **4100** may include airframe **4202** with a plurality of high-level systems **4204** and interior **4206**. Examples of high-level systems **4204** include one or more of propulsion system **4208**, electrical system **4210**, hydraulic system **4212** and environmental system **4214**. Any number of other systems may be included. Although an aerospace example is shown, the principles disclosed herein may be applied to other industries, such as the automotive industry. Accordingly, in addition to aircraft **4200**, the principles disclosed herein may apply to other vehicles, e.g., land vehicles, marine vehicles, space vehicles, etc.

[0130] The disclosed methods **100**, **500**, **600**, **1000**, **1100**, **1200**, **1300**, **1400**, **1500**, **1600**, **1900**, **2000**, **2200**, **2500**, **2800**, **2900**, **3600**, **3700**, **3800** and systems **3900**, **4000** for reinforcing additively manufactured structures **200** may be employed during any one or more of the stages of the manufacturing and service method **4100**. For example, components or subassemblies corresponding to component and subassembly manufacturing (block **4106**) may be fabricated or manufactured in a manner similar to components or subassemblies produced while aircraft **4200** is in service (block **4112**). Also, one or more examples of the system(s), method(s) or combination thereof may be utilized during production stages (block **4106** and block **4108**), for example, by substantially expediting assembly of or reducing the cost of aircraft **4200**. Similarly, one or more examples of the system or method realizations or a combination thereof, may be utilized, for example and without limitation, while aircraft **4200** is in service (block **4112**) and/or during maintenance and service (block **4114**).

[0131] The described features, advantages and characteristics of one example may be combined in any suitable manner in one or more other examples. One skilled in the relevant art will recognize that the examples described herein may be practiced without one or more of the specific features or advantages of a particular example. In other instances, additional features and advantages may be recognized in certain examples that may not be present in all examples. Furthermore, although various examples of the methods **100**, **500**, **600**, **1000**, **1100**, **1200**, **1300**, **1400**, **1500**, **1600**, **1900**, **2000**, **2200**, **2500**, **2800**, **2900**, **3600**, **3700**, **3800** and systems **3900**, **4000** for reinforcing additively manufactured structures **200** have been shown and described, modifications may occur to those skilled in the art upon reading the specification. The present application includes such modifications and is limited only by the scope of the claims.

Claims

1. A method for reinforcing an additively manufactured structure, the method comprising: draping a textile substrate comprising one or more conductive fibers over at least a portion of a surface of the additively manufactured structure, the additively manufactured structure comprising a thermoplastic matrix material; and applying an electric current to at least one conductive fiber of the one or more conductive fibers to heat the thermoplastic matrix material of the additively

- manufactured structure for consolidation of the textile substrate with the thermoplastic matrix material to form a thermoplastic composite on the surface.
2. The method of claim 1 wherein the applying of the electric current to the at least one conductive fiber on the surface of the additively manufactured structure forms a fiber-reinforced additively manufactured structure.
3. The method of claim 1 wherein the additively manufactured structure comprises an outer portion and at least the outer portion comprises the thermoplastic matrix material.
4. The method of claim 1 wherein the additively manufactured structure comprises an inner portion and the inner portion comprises another material different from the thermoplastic matrix material.
- 5-10. (canceled)
11. The method of claim 1 wherein the textile substrate comprises a plurality of the conductive fibers arranged in a first direction and a plurality of non-conductive fibers arranged in a second direction that crosses the first direction.
- 12-13. (canceled)
14. The method of claim 1 wherein the textile substrate comprises reinforcement fibers pre-impregnated with a resin matrix material.
15. (canceled)
16. The method of claim 1, further comprising: cutting the textile substrate from a textile fabric sheet such that the textile substrate is tailored to fit the surface of the additively manufactured structure or at least the portion of the surface over which the textile substrate is draped.
17. (canceled)
18. The method of claim 1 wherein the one or more conductive fibers comprise at least one of one or more carbon fibers, one or more aramid fibers, one or more metal fibers, one or more copper fibers and one or more aluminum fibers.
19. The method of claim 1 wherein at least one of the one or more conductive fibers comprises an insulation coating.
20. The method of claim 1, further comprising: tacking the textile substrate to the surface of the additively manufactured structure in conjunction with the draping of the textile substrate.
21. The method of claim 1, the draping of the textile substrate comprising: wrapping the textile substrate around the surface of the additively manufactured structure; and tacking the textile substrate to the surface of the additively manufactured structure in conjunction with the wrapping.
22. (canceled)
23. The method of claim 1, further comprising: applying the electric current to additional conductive fibers of the one or more conductive fibers to further heat the thermoplastic matrix material of the additively manufactured structure for further consolidation of the textile substrate with the thermoplastic matrix material of the additively manufactured structure to further form the thermoplastic composite on the surface.
24. (canceled)
25. The method of claim 1 wherein the consolidation of the textile substrate with the thermoplastic matrix material is based at least in part on the electric current being applied to the at least one conductive fiber for a predetermined time, the method further comprising: monitoring a time the electric current was applied to the at least one conductive fiber; and removing the electric current from the at least one conductive fiber after the time the electric current was applied exceeds the predetermined time.
26. (canceled)
27. The method of claim 1 wherein the consolidation of the textile substrate with the thermoplastic matrix material is based at least in part on applying a compressive force to the textile substrate, the method further comprising: using at least one of a weighted cover over the textile substrate and a vacuum bagging subsystem.
28. The method of claim 1, further comprising: removing the electric current from the at least one

conductive fiber.

29. The method of claim 1, further comprising: curing the thermoplastic composite on the surface of the additively manufactured structure to form a fiber-reinforced additively manufactured structure.

30-31. (canceled)

32. The method of claim 1 wherein the textile substrate is cut from a first textile fabric sheet such that the textile substrate is tailored to fit a first portion of the surface of the additively manufactured structure, the method further comprising: cutting a second textile substrate comprising one or more second conductive fibers from the first textile fabric sheet or a second textile fabric sheet such that the second textile substrate is tailored to fit a second portion of the surface; draping the second textile substrate over the second portion of the surface; and applying the electric current to at least one second conductive fiber of the one or more second conductive fibers to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the second textile substrate with the thermoplastic matrix material to form a second thermoplastic composite on the surface.

33. (canceled)

34. The method of claim 1, further comprising: adding the one or more conductive fibers to a carrier substrate to form the textile substrate, the one or more conductive fibers forming one or more paths distributed about the textile substrate.

35-44. (canceled)

45. A method for reinforcing an additively manufactured structure, the method comprising: adding a first conductive fiber to a fiber reinforcement fabric substrate to form a textile substrate, the first conductive fiber forming a first path distributed about at least a first area of the textile substrate; cutting the textile substrate to form a tailored fabric substrate that fits a surface of the additively manufactured structure, the additively manufactured structure comprising a thermoplastic matrix material; wrapping the tailored fabric substrate around the surface; and applying an electric current to the first conductive fiber to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the tailored fabric substrate with the thermoplastic matrix material to form a first thermoplastic composite on the surface and thereby form a fiber-reinforced additively manufactured structure.

46-67. (canceled)

68. A system for reinforcing an additively manufactured structure, the system comprising: a fabrication tool set; an electrical power source; and a control device in operative communication with the electrical power source to selectively apply and remove electric current from one or more conductive fibers of a textile substrate, and wherein the textile substrate is configured for draping over at least a portion of a surface of the additively manufactured structure, the additively manufactured structure comprising a thermoplastic matrix material, and wherein the control device and the electrical power source are configured to apply the electric current to at least one conductive fiber of the one or more conductive fibers to heat the thermoplastic matrix material of the additively manufactured structure for consolidation of the textile substrate with the thermoplastic matrix material to form a thermoplastic composite on the surface.

69-95. (canceled)
